Module 2: Statistical inference (I)

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Outline

This module we will review

- Basics of probability
- Fundamental concepts in inference

Probability distributions

- In statistics, we try to draw conclusions about a larger population from a sample of observations.
- We use mathematical models to capture probabilistic behavior of a population.
- This behavior is modeled using probability distributions.

Density/Distribution functions

Definition (Cumulative Distribution Function)

$$F_X(x) = P(X \leq x) \quad \forall x \in \mathbb{R}$$

Density/Distribution functions (cont'd)

Definition (Probability Mass Function)

For a discrete RV, the probability mass function (PMF) is:

$$f_X(x) = P(X = x) \quad \forall x \in \mathbb{R}$$

Definition (Probability Density Function)

For a continuous RV, the probability density function (PDF) is:

$$f_X(x) = \frac{\partial}{\partial t} F(t)\bigg|_{t=x}$$

So $F_X(x) = \int_{-\infty}^{x} f_X(t) dt \forall x \in \mathbb{R}$.

Note that $f_X \geq 0$ for $\forall x$, and thus F_X is an increasing function.

Expectation and Variance

Definition (Expectation)

A measure of central tendancy (a weighted average of the values of X)

$$E[X] = \sum_{x \in S} xP(X = x) \text{ for discrete RV taking values from } S$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \text{ for continuous RV}$$

Definition (Variance)

A measure of the spread of a distribution

$$\mathrm{Var}(X) = \sum_{x \in S} (x - E[X])^2 P(X = x) \text{ for discrete RV}$$

$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x - E[X])^2 f_X(x) dx \text{ for continuous RV}$$

Discreate random variable

A discrete random variable has a countable number of possible values.

Bernoulli and Binomial random variable

- Consider the event of flipping a (possibly unfair) coin.
- $Y \in \{0, 1\}$ represents success and failure.
- Suppose we only flip the coin once,
  - We can express $P(Y = 1) = p$ and $P(Y = 0) = 1 - p$
- Bernoulli distribution

$$P(Y = y) = p^y (1 - p)^{1-y} \quad \text{for} \quad y = 0, 1$$

- If we flip the coin n times,
- Binomial distribution

$$P(Y = y) = \binom{n}{y} p^y (1 - p)^{n-y} \quad \text{for} \quad y = 0, 1, \ldots, n$$

Binomial distributions with different values of n and p

If $Y \sim \text{Binomial}(n, p)$, then $\mathrm{E}(Y) = np$ and $\mathrm{SD}(Y) = \sqrt{np(1-p)}$.

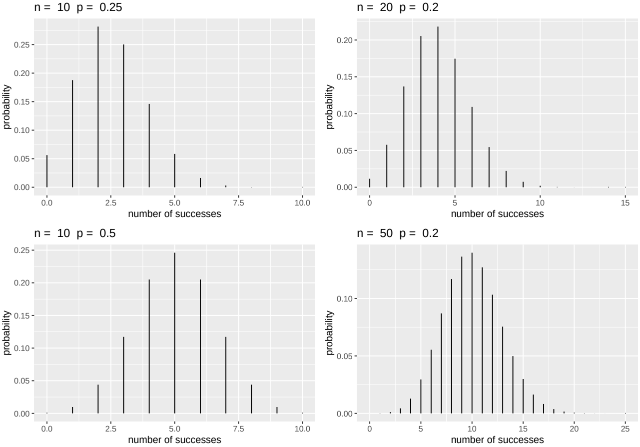

Figure 1: Binomial distributions with different values of n and p.

How to generate in R?

All common distributions have four functions in R:

- Density
  `dbinom(x, size, prob)`
- Distribution function
  `pbinom(q, size, prob)`
- Quantile function
  `qbinom(p, size, prob)`
- Random generaation
  `rbinom(n, size, prob)`

Not sure? Using `?` with any of the four functions, e.g. `?qbinom`

Example of binomial distribution computing

Question: While taking a multiple choice test, a student encountered 10 problems where she ended up completely guessing, randomly selecting one of the four options. What is the chance that she got exactly 2 of the 10 correct?

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Answer: Knowing that the student randomly selected her answers, we assume she has a 25% chance of a correct response.

$$P(Y = 2) = \left(\begin{array}{c} 10 \\ 2 \end{array} \right) (.25)^2 (.75)^8 = 0.282$$

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R computing:

```
dbinom(2, size = 10, prob = .25)
```

```
## [1] 0.2815676
```

Continuous random variable

A continuous random variable can take on an uncountably infinite number of values. Given a pdf $f(y)$,

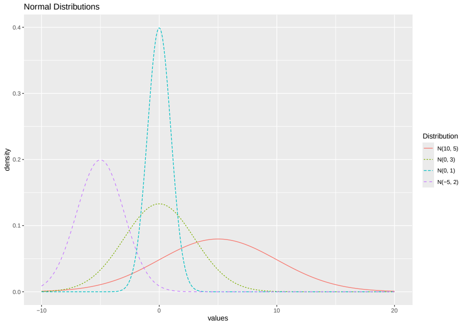
$$P(a \leq Y \leq b) = \int_a^b f(y)dy$$

Properties:

- $\int_{-\infty}^{\infty} f(y)dy = 1$.
- For any value y, $P(Y = y) = \int_y^y f(y)dy = 0$. $P(y < Y) = P(y \leq Y)$.

Example of Continuous Distribution (Normal)

- The normal distribution is a very important distribution because:
  - A lot of things look normal
  - Analytically tractable
  - Central limit theorem
- $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$
- Characterized by mean, μ, and variance, σ^2.



How to Generate Samples from Normal Distribution

The following commands are for a normal random variable with mean μ and variance σ^2, that is, $X \sim N(\mu, \sigma^2)$,

- To calculate the probability density function at a value x, `dnorm(x,mu,sigma)`
- To calculate the cumulative distribution function at a value x, `pnorm(x,mu,sigma)`
- To generate a size m sample from the normal distribution, `rnorm(m,mu,sigma)`
- Note that the third argument is the **square root of the variance**, this is because the R function for normal distribution asks for the standard deviation, which is defined as the square root of the variance

Some probability distributions in R

Continuous

- Normal (`?rnorm`)
- Uniform (`?runif`)
- Beta (`?rbeta`)
- Chi-sq (`?rchisq`)
- Exponential (`?rexp`)
- t (`rt`)
- F (`?rf`)
- Logistic (`?rlogis`)
- Lognormal (`?rlnorm`)

Discrete

- Poisson (`?rpois`)
- Binomial (`?rbinom`)
- Geometric (`?rgeom`)
- Negative Binomial (`?rnbinom`)
- Multinomial (`?rmultinom`)

Empirical vs. Theoretical CDF

In statistics, an empirical distribution function is the distribution function associated with the empirical measure of a sample.

- Theoretical CDF

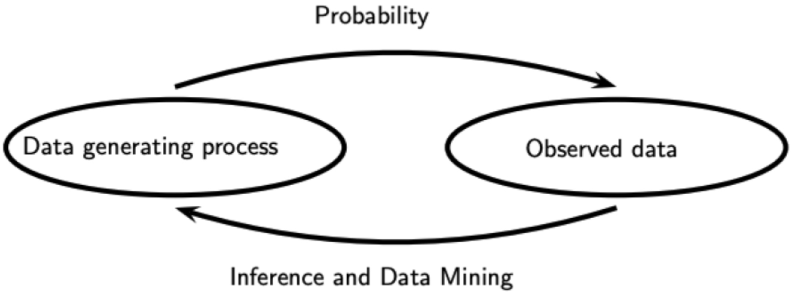
$$F_X(k) = \Pr(X \leq k)$$

- Empirical CDF

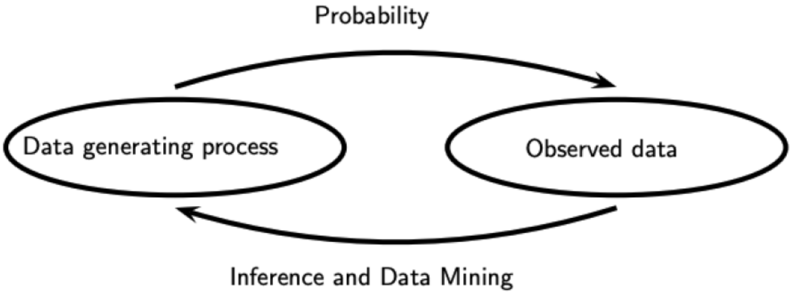
$$\hat{F}_n(k) = \frac{\text{number of elements in the sample } \leq k}{n} = \frac{1}{n}\sum_{i=1}^{n} I_{X_i \leq k}$$

where $X_1, \ldots, X_n$ make up some random sample from the underlying distribution.

Probability and inference



Probability and inference



- Probability: Given a data generating process, what are the properties of the outcomes?
- Statistical inference: Given the outcomes, what can we say about the process that generated the data?

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$$\mathfrak{F} = \{f(x;\theta) : \theta \in \Theta\}$$

  where θ is an unknown parameter (or vector of parameters) that can take values in the parameter space Θ.
  - e.g. Normal distribution, a 2-parameter model with density as $f(x;\mu,\sigma)$

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  - e.g. Normal distribution, a 2-parameter model with density as $f(x; \mu, \sigma)$

- Nonparametric model: a set $\mathfrak{F}$ that cannot be parameterized by a finite number of parameters
  - e.g. $\mathfrak{F}_{\mathrm{ALL}} = \{\text{ all } \mathrm{CDF}'s\}$ is nonparametric.

Frequentist and Bayesian

- Frequentist: statistical methods with guaranteed frequency behavior
- Bayesian: statistical methods for using data to update beliefs

Point estimation

- Providing a single “best guess” of some quantity of interest
- Notations
  - Parameter θ: fixed, unknown quantity
  - Point estimator $\hat{\theta}$: depends on data, random variable

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Definition (Point estimator)

Let $X_1, \ldots, X_n$ be n IID data points from some distribution F. A point estimator $\hat{\theta}_n$ of a parameter θ is some function of $X_1, \ldots, X_n$:

$$\widehat{\theta}_n = g\left(X_1, \ldots, X_n\right)$$

- What is a good point estimate?

MSE

- Definition:
$$\mathrm{MSE} = \mathbb{E}_\theta \left(\widehat{\theta}_n - \theta\right)^2$$
- No uniformly best estimator in terms of MSE
- It is NOT possible to have an estimator that is uniformly the best.

Bias and Variance

- Bias

$$\text{bias}\left(\widehat{\theta}_n\right) = \mathbb{E}_\theta\left(\widehat{\theta}_n\right) - \theta$$

- Variance

$$\mathrm{Var}\left(\widehat{\theta}_n\right) = \mathbb{E}_\theta\left(\widehat{\theta}_n - \mathbb{E}\theta\right)^2$$

- Theorem

$$MSE = bias^2 + Var$$

Unbiasedness

- Definition

$$\text{bias}\left(\widehat{\theta}_n\right) = \mathbb{E}_\theta\left(\widehat{\theta}_n\right) - \theta = 0$$

- Unbiasedness is a small sample (finite sample) property
- An unbiased estimator may not exist
- An unbiased estimator is not necessarily a good estimator

Consistency

- Definition

$$\widehat{\theta}_n \xrightarrow{\mathrm{P}} \theta$$

- It is possible to be unbiased but not consistent.
- It is possible to be consistent but not unbiased.

Asypototic unbiasedness

- Definition

$$\text{bias}\left(\widehat{\theta}_n\right) = \mathbb{E}_\theta\left(\widehat{\theta}_n\right) - \theta \to 0, \text{ as } n \to \infty$$

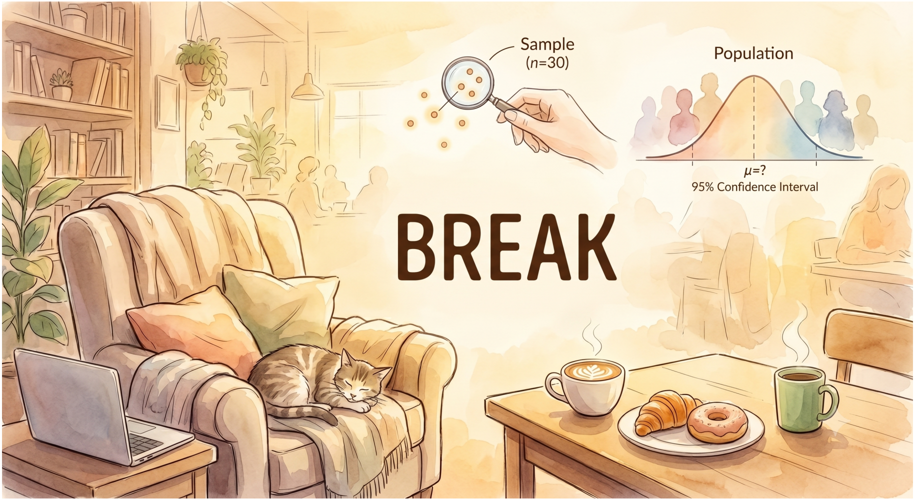
- It is possible to be asypototically unbiased but not consistent.
- It is possible to be consistent but NOT asymptotically unbiased.
- Sufficient conditions: $MSE \to 0$.

Resources

This tutorial is based on

- Havard Biostatistics Summer Pre Course [link]
- "Beyond Multiple Linear Regression" by Paul Roback and Julie Legler [link]

Break



Outline

This module we will review

- Basics of parametric inference
- Methods for generating parametric estimators
- Maximum likelihood estimators
- Delta method
- Optimization method for finding MLE in R (Newton-Raphson, EM algorithm)

Parametric inference

Definition (Parametric models)

$$\mathfrak{F} = \{f(x;\theta) : \theta \in \Theta\}$$

where the $\Theta \subset \mathbb{R}^k$ is the parameter space and $\theta = (\theta_1, \ldots, \theta_k)$ is the parameter.

Goal of parametric inference

- estimate the parametric θ (assume we known the form of the density).

Parameter of interest and nuisance parameter

Often, we are interested in estimating some function $T(\theta)$.

For example, if $X \sim N(\mu, \sigma^2)$, then

- Parameters: $\theta = (\mu, \sigma)$
- Parameter space: $\Theta = \{(\mu, \sigma) : \mu \in \mathbb{R}, \sigma > 0\}$

If the goal is to estimate the μ then

- Parameter of interest: $T(\theta) = \mu$
- Nuisance parameter: σ

Methods for generating parametric estimators

1. Method of moments
2. Maximum likelihood

Method of moments

Definitions

- $\mathbb{E}(X^k)$ is the k^{th} (theoretical) moment of the distribution, for $k = 1, 2, ...$
- $M_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$ is the k^{th} sample moment, for $k = 1, 2, ...$

Steps to find MoM

The basic idea behind this form of the method is to:

- Equate the first sample moment about the origin $M_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$ to the first theoretical moment $\mathbb{E}(X)$.
- Equate the second sample moment about the origin $M_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$ to the second theoretical moment $\mathbb{E}(X^2)$.
- Continue equating sample moments about the origin until you have as many equations as you have parameters.
- Solve for the parameters

Example of MoM (Bernoulli)

Let $X_1, ..., X_n$ be Bernoulli random variables with parameter p. What is the method of moments estimator of p?

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Let $X_1, ..., X_n$ be Bernoulli random variables with parameter p. What is the method of moments estimator of p?

Only one parameter p, so we only need to equate the first moment.

$\mathbb{E}(X_i) = p = M_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$.

So,

$$\hat{p}_{MoM} = \bar{X}$$

Example of MoM (Normal)

Let $X_1, ..., X_n$ be normal random variables with mean μ and variance σ^2. What are the method of moments estimators of the mean μ and variance σ^2 ?

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Let $X_1, ..., X_n$ be normal random variables with mean μ and variance σ^2. What are the method of moments estimators of the mean μ and variance σ^2 ?

Two parameters, so we need to equate the first and the second moment.

$E(X_i) = \mu = M_1, E(X_i^2) = \sigma^2 + \mu^2 = M_2$.

So,

$$\hat{\mu}_{MoM} = \bar{X}, \hat{\sigma}^2_{MoM} = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2.$$

Asymptotic properties

Under mild regularity conditions, MoM estimators are

- *Consistent* $\rightarrow$ converge to the true value in probability as $n \rightarrow \infty$, i.e.

$$\lim_{n\rightarrow\infty} P(|\hat{\theta} - \theta| \leq \epsilon) = 1 \quad \forall \epsilon > 0$$

- *Asymptotically Normal* $\rightarrow \sqrt{n}(\hat{\theta} - \theta) \sim N\left(0, \sigma^2\right)$ for large n
- However, they are usually **NOT** *Asymptotically Efficient*

Maximum likelihood

- Parametric model: $f(x;\theta)$, $X_1, \ldots, X_n$ iid
- Likelihood function

$$\mathcal{L}_n(\theta) = \prod_{i=1}^{n} f(X_i;\theta)$$

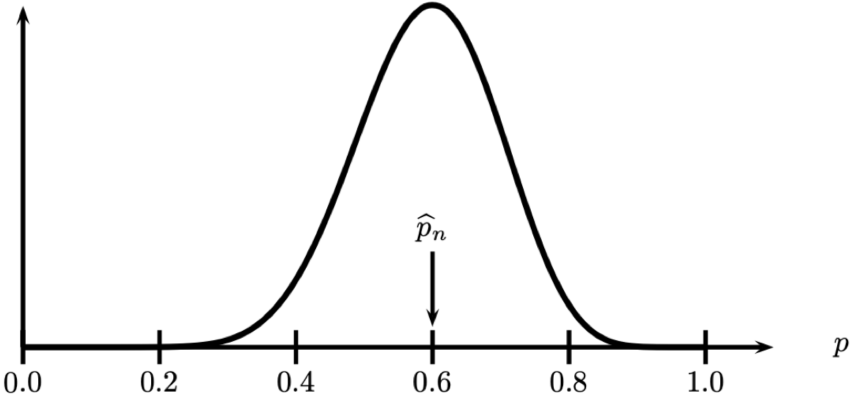
- The log-likelihood function

$$\ell_n(\theta) = \log \mathcal{L}_n(\theta) = \sum_{i=1}^{n} \log f(X_i;\theta)$$

- The maximum likelihood estimator (MLE)

$$\hat{\theta}_{MLE} = \arg\max_{\theta} \mathcal{L}(\theta)$$

An example of MLE



Likelihood function for Bernoulli with $n = 20$ and $\sum_{i=1}^{n} X_i = 12$. The MLE is $\widehat{p}_n = 12/20 = 0.6$.

Steps to find the MLE

1. Write out the likelihood

$$\mathcal{L}(\theta) = f\left(X_1, \ldots, X_n; \theta\right)$$

2. Simplify the log likelihood

$$\ell(\theta) = \log \mathcal{L}(\theta)$$

3. Take the derivative of $\ell(\theta)$ with respect to the parameter of interest, θ
   Set $= 0$
4. Solve for θ (get $\hat{\theta}_{MLE}$)
5. Check that $\hat{\theta}_{MLE}$ is a maximum $\left(\frac{\partial^2}{\partial\theta^2}\ell(\theta) < 0\right)$

Exercise

Suppose we have an iid sample $\{X_1, \ldots, X_n\}$ with $X_i \sim$ Bernoulli(p). Find the MLE for p.

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where $S = \sum_i X_i$

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2. Log-likelihood

$$\ell_n(p) = S \log p + (n-S)\log(1-p)$$

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2. Log-likelihood

$$\ell_n(p) = S \log p + (n - S) \log(1-p)$$

3. MLE

$$\ell_n'(p) = 0$$

The MLE is $\widehat{p}_n = S/n$.

Asymptotics of MLE

Under mild regularity conditions, MLEs are

- *Consistent* $\rightarrow$ converge to the true value in probability as $n \rightarrow \infty$, i.e.

$$\lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta| \leq \epsilon) = 1 \quad \forall \epsilon > 0$$

- *Asymptotically normal* $\rightarrow \sqrt{n}(\hat{\theta} - \theta) \sim N\left(0, \sigma^2\right)$ for large n
- *Asymptotically efficient*
- *equivariant* $\rightarrow$ if $\hat{\theta}$ is the MLE for θ then $g(\hat{\theta})$ is the MLE for $g(\theta)$

Asymptotic Efficiency

Cramér–Rao bound

The variance of any *unbiased* estimator $\hat{\theta}$ of θ is bounded by the reciprocal of the Fisher information $I(\theta)$:

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)},$$

where $I(\theta) = n\mathbb{E}\left[\left(\frac{\partial \ell}{\partial \theta}\right)^2\right]$.

Both MoM estimators are asymptotically unbiased, but MLE estimators achieves the CR lower bound.

MLE in R

Sometimes, there is no closed-form solution, so we need to use optimization methods to find the maximum of the log-likelihood.

- `optim()` find values of some parameters that **minimizes** some function.
- Newton-Raphson
- EM-algorithm

Example using optim()

```r
set.seed(42) # For reproducibility
sample_data <- rbinom(1000, size = 1, prob = 0.3) # Assuming success probability of 0.3

# Log-likelihood function for Bernoulli distribution
log_likelihood_bernoulli <- function(p, data) {
  n <- length(data)
  log_likelihood <- sum(data * log(p) + (1 - data) * log(1 - p))
  return(-log_likelihood) # Negative to be used with optimization functions (minimization)
}

# Initial parameter value for optimization (probability of success)
initial_param <- 0.8

# Find MLE using optim
result <- optim(
  par = initial_param, fn = log_likelihood_bernoulli,
  data = sample_data, method = "Brent", lower = 0, upper = 1
) # If bounds aren't provided, use method="BFGS"


# MLE estimate of p
mle_p <- result$par

# Print the result
cat("MLE of p:", mle_p, "\n")
```

```
## MLE of p: 0.293
```

Newton-Raphson

Derivative of the log-likelihood around θ^j :

$$0 = \ell'(\widehat{\theta}) \approx \ell'\left(\theta^j\right) + \left(\widehat{\theta} - \theta^j\right)\ell''\left(\theta^j\right)$$

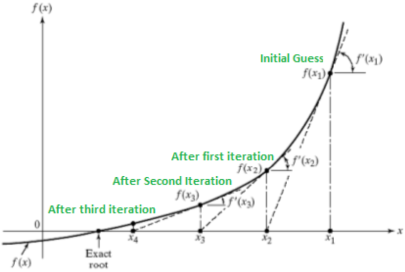
Solving for $\widehat{\theta}$ gives

$$\widehat{\theta} \approx \theta^j - \frac{\ell'\left(\theta^j\right)}{\ell''\left(\theta^j\right)}.$$

This suggests the following iterative scheme:

$$\widehat{\theta}^{j+1} = \theta^j - \frac{\ell'\left(\theta^j\right)}{\ell''\left(\theta^j\right)}$$

Illustration

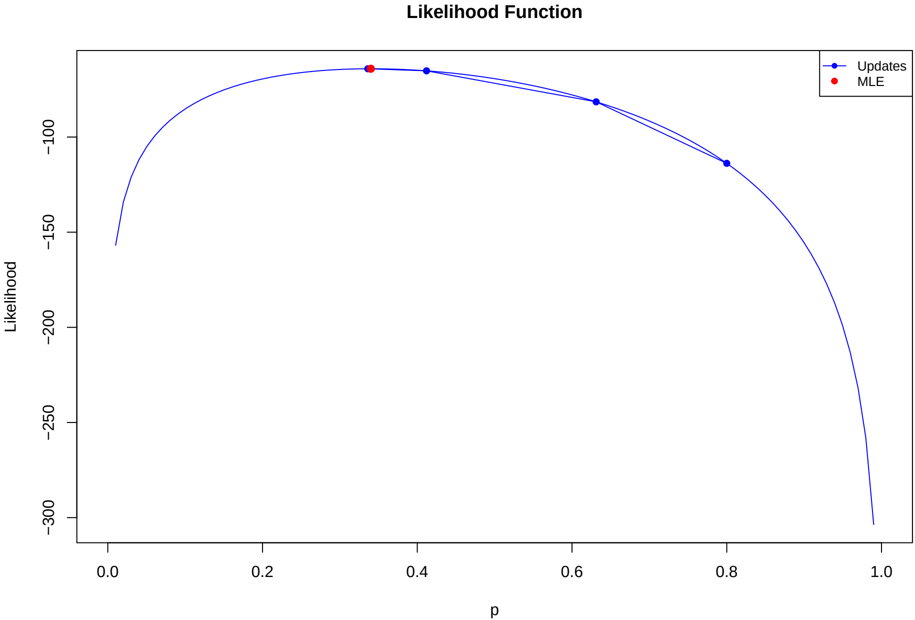


NR algorithm in R

```r
# First derivative of the log-likelihood function
log_likelihood_bernoulli_prime <- function(p, data) {
  n <- length(data)
  d_log_likelihood <- sum(data / p - (1 - data) / (1 - p))
  return(-d_log_likelihood) # Negative to be used with optimization functions (minimization)
}
# Second derivative of the log-likelihood function
log_likelihood_bernoulli_double_prime <- function(p, data) {
  n <- length(data)
  dd_log_likelihood <- sum(-data / p^2 - (1 - data) / (1 - p)^2)
  return(-dd_log_likelihood) # Negative to be used with optimization functions (minimization)
}
# Initial parameter value for optimization (probability of success)
initial_param <- 0.8
# Newton-Raphson algorithm for optimization
tolerance <- 1e-8
max_iterations <- 1000
p <- initial_param
for (i in 1:max_iterations) {
  p_new <- p - log_likelihood_bernoulli_prime(p, sample_data) /
    log_likelihood_bernoulli_double_prime(p, sample_data)
  if (abs(p_new - p) < tolerance) {
    break
  }
  p <- p_new
}
# Print the result
cat("MLE of p:", p, "\n")
```

```
## MLE of p: 0.293
```

Solution Trajectory



Delta method

Theorem (The Delta Method).

Suppose that

$$\frac{\sqrt{n}\,(Y_n - \mu)}{\sigma} \rightsquigarrow N(0,1)$$

and that g is a differentiable function such that $g'(\mu) \neq 0$. Then

$$\frac{\sqrt{n}\,(g\,(Y_n) - g(\mu))}{|g'(\mu)|\,\sigma} \rightsquigarrow N(0,1).$$

In other words,

$$Y_n \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{implies that} \quad g\,(Y_n) \approx N\left(g(\mu), (g'(\mu))^2\,\frac{\sigma^2}{n}\right).$$

Exercise

Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ and let $\psi = g(p) = \log(p/(1- p)\ )$. Find the MLE of ψ and its asymptotic distribution.

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The MLE of p was derived previously as follows

$$\hat{p}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$$

Now, for $\psi = g(p) = \log\left(\frac{p}{1-p}\right)$, the MLE of ψ is obtained by plugging in the MLE of p:

$$\hat{\psi}_{\text{MLE}} = g(\hat{p}_{\text{MLE}}) = \log\left(\frac{\hat{p}_{\text{MLE}}}{1-\hat{p}_{\text{MLE}}}\right) = \log\left(\frac{\bar{X}}{1-\bar{X}}\right)$$

Exercise

By the Central Limit Theorem, as $n \to \infty$:

$$\sqrt{n}(\bar{X} - E[X_i]) \xrightarrow{D} N(0, \mathrm{Var}(X_i))$$

Therefore,

$$\sqrt{n}(\hat{p}_{\mathrm{MLE}} - p) \xrightarrow{D} N(0, p(1-p))$$

To apply delta method, we need to find the derivative of $g(p)$:

$$g'(p) = \frac{1}{p(1-p)}$$

and

$$[g'(p)]^2\sigma^2 = \left(\frac{1}{p(1-p)}\right)^2 \cdot p(1-p) = \frac{1}{p^2(1-p)^2} \cdot p(1-p) = \frac{1}{p(1-p)}$$

Therefore,

$$\sqrt{n}(\hat{\psi}_{\mathrm{MLE}} - \psi) \xrightarrow{D} N\left(0, \frac{1}{p(1-p)}\right)$$

Resources

This tutorial is based on

- Harvard Biostatistics Summer Pre Course [link]
- "All of Statistics" by Larry A. Wasserman [link]

Exercises

Available on course website.